

$$x + 1 = y + 1$$

$$\underline{\underline{\text{plus } x \ 1}} = \underline{\underline{\text{plus } y \ 1}}$$

$$\underbrace{\text{plus } 1 \ x}_{\text{S } x} = \text{plus } 1 \ y$$

$$\text{S } x = \text{S } y$$

```

Fixpoint eqb (n m : nat) : bool :=
  match n with
  | 0 => match m with
        | 0 => true
        | S m' => false
      end
  | S n' => match m with
            | 0 => false
            | S m' => eqb n' m'
          end
  end.

```

```

Fixpoint plus (n : nat) (m : nat) : nat :=
  match n with
  | 0 => m
  | S n' => S (plus n' m)
  end.

```

to show $\forall n m. \text{double } n = \text{double } m \rightarrow n = m$

Let n, m be natural numbers.

proof by induction on n

```
Fixpoint double (n:nat) :=  
  match n with  
  | 0 => 0  
  | S n' => S (S (double n'))  
end.
```

case $n=0$ t.s. $\forall m. \text{double } 0 = \text{double } m \rightarrow 0 = m$
 $\Leftrightarrow 0 = \text{double } m \rightarrow 0 = m$

Let m be a natural number
case distinction on m t.s.

$m=0$ Assume $0 = \text{double } 0$ and show that $0 = 0$ ✓

$m = S m'$ t.s. $0 = \text{double } (S m') \rightarrow 0 = S m'$

Assume $0 = S(S(\text{double } m'))$ we show that $0 = S m'$

by disjointness of constructors this gives a contradiction.

case $n = S n'$ with $\forall m. [\text{double } n' = \text{double } m \rightarrow n' = m \text{ (I.H.)}]$

t.s. $\forall m. \text{double } (S n') = \text{double } m \rightarrow S n' = m$

case distinction on m .

$m = 0$ † s. $\underbrace{\text{double}(S_{n'})}_{S(S(\text{double } n'))} = \underbrace{\text{double } 0}_0 \rightarrow S_{n'} = 0$

(same as previous case)

case $m = S_{m'}$

† s. $\underbrace{\text{double}(S_{n'})}_{S(S(\text{double } n'))} = \underbrace{\text{double}(S_{m'})}_{S(S(\text{double } m'))} \rightarrow S_{n'} = S_{m'}$

double m

$$P_1(n) := \text{double } m = \text{double } n \rightarrow n = m$$

$$P_2(n) := \forall m. \text{double } n = \text{double } m \\ \rightarrow n = m$$

- General principle to show that $\forall n \in \mathbb{N}. P(n)$
 - show $P(0)$
 - show that for any n' if $P(n')$ holds, then so does $P(S\ n')$